

Data Analyst Capstone Project

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OUTLINE



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EXECUTIVE SUMMARY



Point 1 : Analyzed job postings and developer survey datasets to identify technology and job market trends.

Point 2 : Explored current and future demand for programming languages, databases, platforms, and web frameworks.

Sub Points

Job market analysis using job postings

Technology trend analysis using developer survey

Demographic analysis of developers

Point 3 : Performed data cleaning, exploratory data analysis, and visualization using Python.

Point 4 : Built interactive dashboards using IBM Cognos Analytics.

Point 5 : Generated actionable insights and recommendations for technology adoption and career planning.

INTRODUCTION



- **Purpose of the report, Target Audience & Value of the analysis**
- **Point 1:** Analyze current technology usage and future technology trends.
- **Point 2:** Identify the most in-demand programming languages, databases, platforms, and web frameworks.
- **Point 3:** Present insights using data visualization and interactive dashboards.
- **Point 4:** Support data-driven decision-making for career planning and technology adoption.

Sub Point 1: Target Audience: Students, developers, recruiters, and business professionals.

Sub Point 2: Tools Used: Python, SQL, IBM Cognos Analytics, and Microsoft PowerPoint.

METHODOLOGY

Data Sources, Data Collection & Data Wrangling Steps

Point 1: Collected data from job postings and the Stack Overflow Developer Survey dataset.

Point 2: Imported and explored the datasets using Python and SQL.

Point 3: Performed data cleaning and preprocessing to improve data quality.

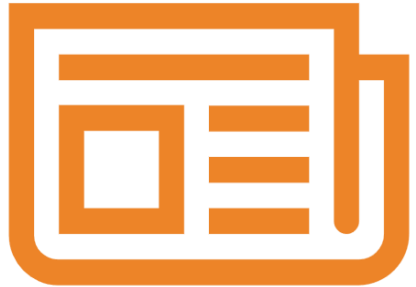
Point 4: Conducted exploratory data analysis and created interactive dashboards using IBM Cognos Analytics.

Sub Point 1

- Data wrangling included handling missing values, removing duplicates, and converting data types.

Sub Point 2

- Visualizations were created using Python libraries and IBM Cognos to identify technology trends and demographic insights.



Results

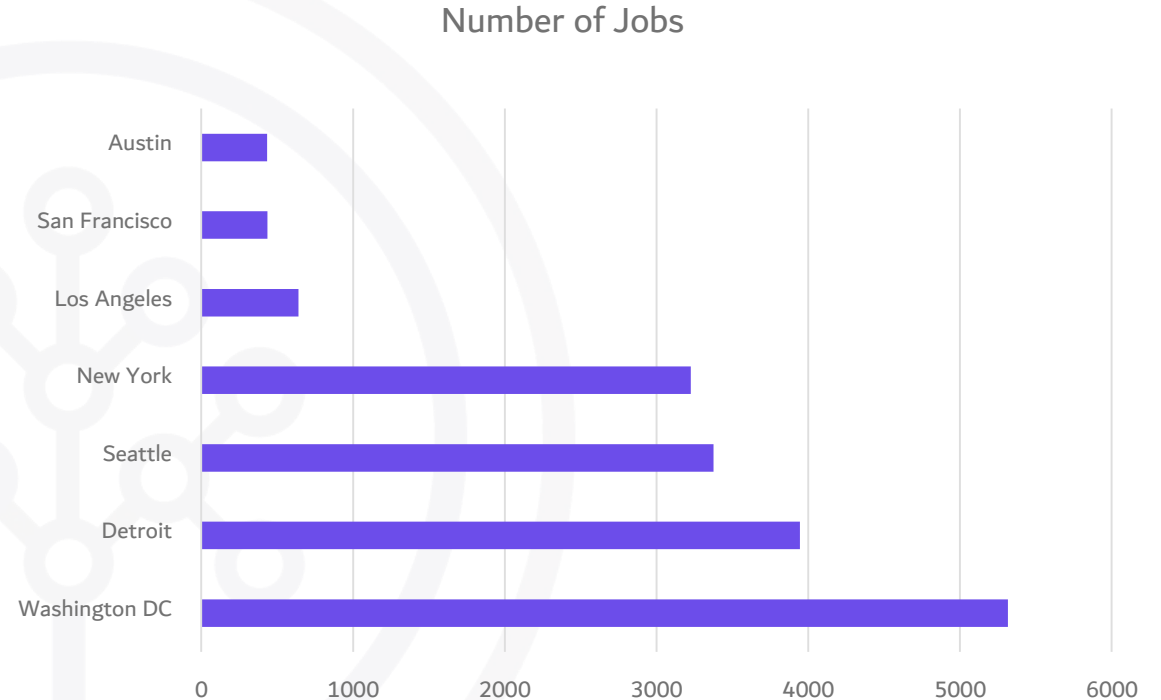


This section presents the key findings of the analysis performed during the capstone project. The following slides summarize the results obtained from job market analysis, technology trends, data exploration, and dashboard visualizations using Python and IBM Cognos Analytics.

The following slides present the key results and insights obtained from the data analysis and dashboard visualizations.

JOB POSTINGS

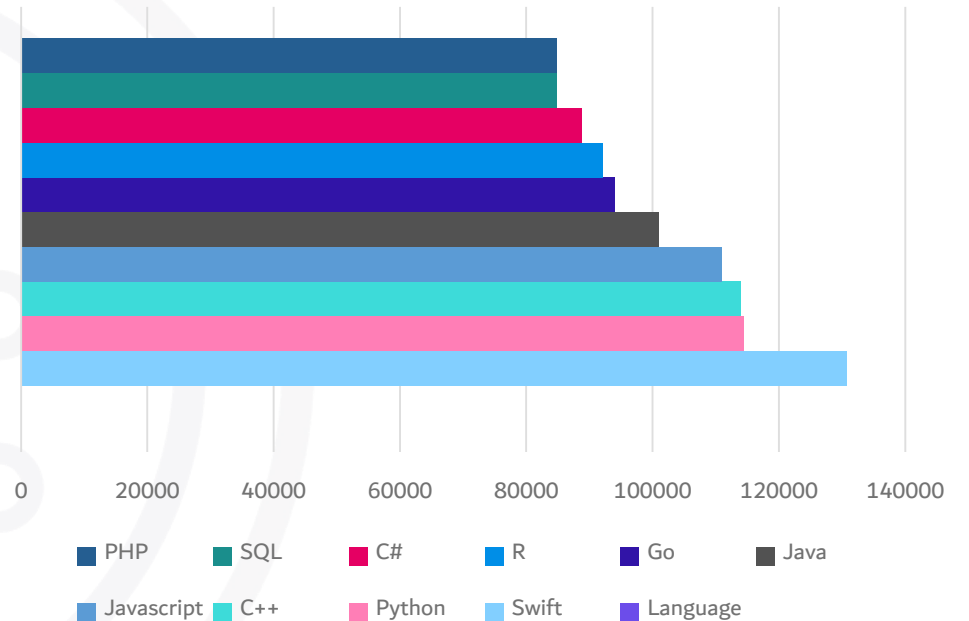
- **Title: Job Postings Analysis**
- **Bullet Points:**
- The chart shows the number of job postings by technology.
- The results are sorted in descending order.
- Technologies with more job postings have higher market demand.



POPULAR LANGUAGES

- The chart compares average salaries for popular programming languages.
- Languages are sorted from highest to lowest salary.
- Higher-paid languages indicate stronger earning potential in the job market

Popular Programming Languages by Salary



PROGRAMMING LANGUAGE TRENDS

Summarize key trends shown in the charts.

Current Year

JavaScript, Python, and SQL dominate as the most widely used languages. Go and Swift show strong mid-tier usage, while PHP and C++ remain relevant in legacy systems.

Next Year

Python and JavaScript are projected to maintain top positions. Demand for Go and Rust is expected to rise due to cloud and systems programming growth, while PHP usage may continue to decline.



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings:

Finding 1: JavaScript and Python consistently rank as the highest-paid and most-used languages across industries.

Finding 2: Go and Swift offer above-average salaries despite lower overall usage, indicating high demand for specialized skills.

Finding 3: Languages like PHP and C++ show lower average salaries, reflecting market saturation.

Implications:

Implication 1: Developers should prioritize learning Python and JavaScript to maximize employability and earning potential.

Implication 2: Investing in Go or Swift skills can lead to higher compensation in niche but growing markets.

Implication 3: Organizations should plan transitions away from lower-paying legacy languages to stay competitive in talent acquisition.



DATABASE TRENDS

Summarize key trends shown in the charts.

Current 10 Years

MySQL, PostgreSQL, and SQLite are the most widely used databases among developers in the current year. Microsoft SQL Server and MongoDB also maintain strong adoption, particularly in enterprise environments, while Oracle continues to play an important role in large organizations despite being less common among individual developers.

Next 10 Years

PostgreSQL and MongoDB are expected to gain wider adoption due to their scalability and cloud support, while MySQL is projected to remain one of the most preferred databases. At the same time, interest in Redis and other cloud-native databases is likely to increase, whereas Oracle adoption may gradually decline in modern application development.



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

Finding 1: MySQL, PostgreSQL, and SQLite are the most widely used databases.

Finding 2: PostgreSQL and MongoDB show strong future growth and increasing developer interest.

Finding 3: Oracle remains important for enterprise applications, while Redis and cloud-native databases are gaining popularity.

Implications

Implication 1: Learning PostgreSQL and MySQL can improve career opportunities in database management.

Implication 2: MongoDB and cloud-native databases are valuable skills for modern application development.

Implication 3: Organizations should adopt scalable and cloud-compatible databases to meet future technology demands.



DASHBOARD TAB 1: Current Technology Usage

Summary

The current technology usage dashboard shows that **JavaScript**, **HTML/CSS**, **SQL**, and **Python** are the most widely used programming

languages among developers. **PostgreSQL**, **Microsoft SQL Server**, and **MySQL**

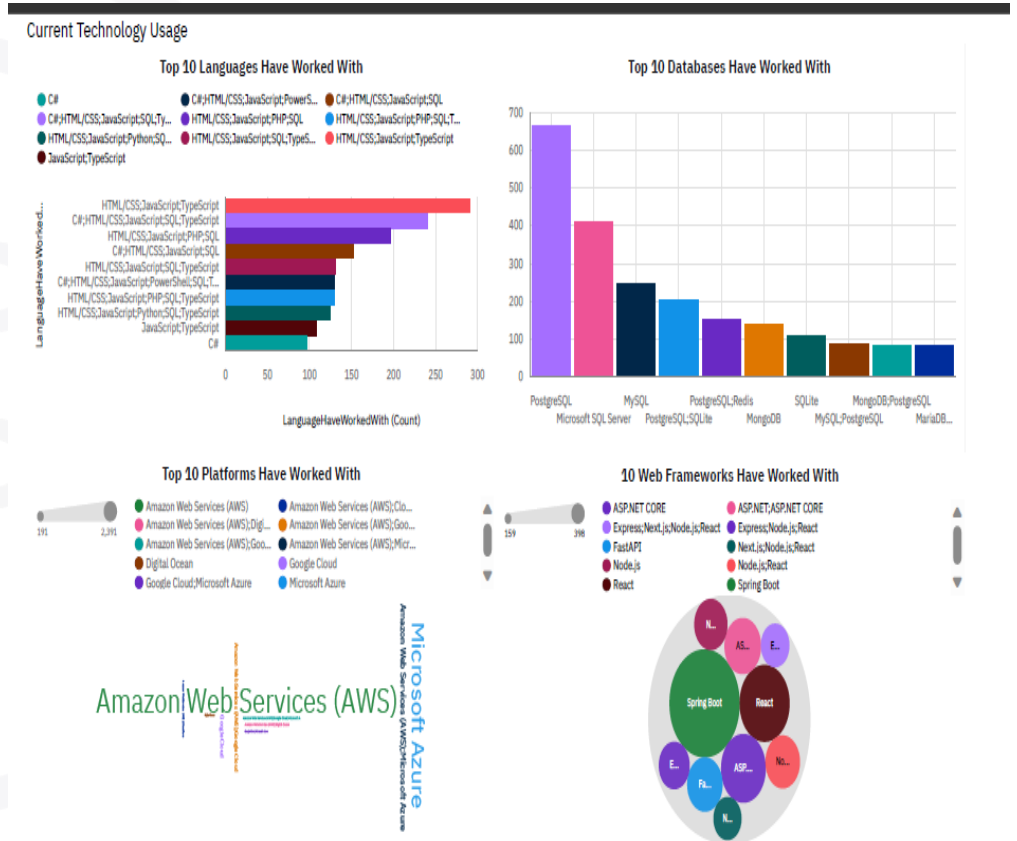
are the leading databases in current use. Among cloud platforms,

Amazon Web Services (AWS) and **Microsoft Azure** dominate the market,

while **Spring Boot**, **React**, and **ASP.NET Core** are the most commonly used

web frameworks. These findings indicate that full-stack development and cloud

technologies continue to play a significant role in modern software development.

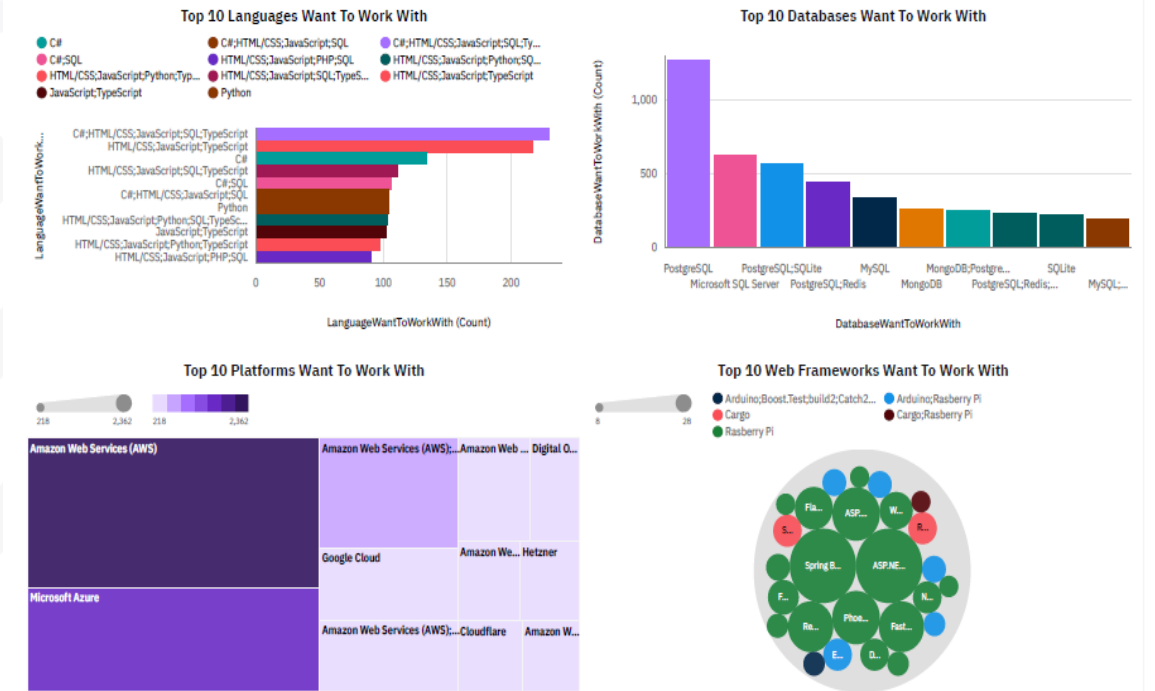


DASHBOARD TAB 2: Future Technology Trends

Summary

The future technology trends dashboard highlights that **Python, JavaScript, and SQL** are expected to remain the most preferred programming languages. **PostgreSQL** continues to be the most desired database, followed by **Microsoft SQL Server** and **MySQL**. Developers also show increasing interest in **AWS, Microsoft Azure, and Google Cloud** for cloud computing. The popularity of **ASP.NET Core, Spring Boot, and React** suggests continued growth in modern web application development.

Future Technology Trends

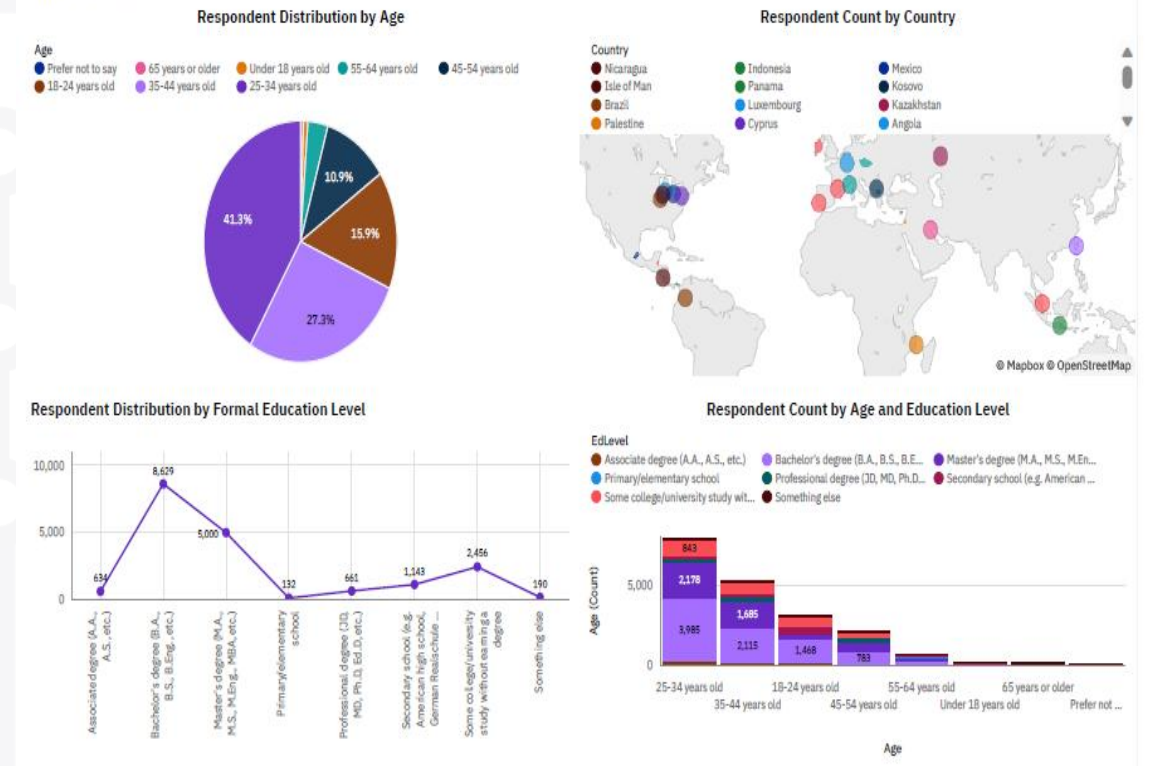


DASHBOARD TAB 3: Demographics

Summary

The demographics dashboard reveals that the majority of respondents belong to the **25–34 years** age group, followed by **35–44 years**. Most respondents hold a **Bachelor's degree**, indicating a highly educated developer community. The country distribution shows participation from multiple regions worldwide, demonstrating the global nature of the survey. The relationship between age and education suggests that the largest proportion of experienced professionals falls within the 25–34 age category.

Demographics.



DISCUSSION



The analysis indicates that Python, JavaScript, SQL, PostgreSQL, AWS, and modern web frameworks such as React and Spring Boot continue to dominate both current usage and future technology preferences. These technologies remain highly valuable for developers seeking career growth and organizations planning future technology adoption. The demographic analysis further shows that the global developer community is largely composed of young, well-educated professionals, highlighting strong demand for modern technical skills.



OVERALL FINDINGS & IMPLICATIONS

Findings

Finding 1: JavaScript, Python, SQL, and PostgreSQL are the most widely used and preferred technologies.

Finding 2: AWS, Microsoft Azure, and Google Cloud continue to dominate cloud platform adoption.

Finding 3: Most survey respondents are aged **25–34 years** and hold a **Bachelor's degree**, representing a skilled developer workforce.

Implications

Implication 1: Developers should focus on Python, JavaScript, SQL, and PostgreSQL to improve career opportunities.

Implication 2: Organizations should invest in cloud technologies such as AWS and Azure to remain competitive.

Implication 3: Continuous learning of modern technologies and cloud platforms is essential to meet future industry demands.



CONCLUSION



Point 1: The analysis identified the current and future trends in programming languages, databases, cloud platforms, and web frameworks.

Point 2: Python, JavaScript, PostgreSQL, and AWS emerged as the most popular and in-demand technologies across the analysis.

Point 3: Interactive dashboards and visualizations provided valuable insights into developer preferences, technology adoption, and demographic patterns.

Point 4: The project demonstrates how data analytics and visualization can support informed career decisions and technology adoption for individuals and organizations.

APPENDIX



Additional Charts

- Histogram of Yearly Compensation
- Scatter Plot of Compensation vs. Job Satisfaction
- Bubble Plot of Technology Preferences by Age
- Pie Charts for Programming Languages and Databases

Additional Tables

- Top 10 Programming Languages
- Top 10 Databases
- Summary Statistics of Survey Data

Supporting Materials

- IBM Cognos Dashboard Screenshots
- Python Analysis Outputs
- Job Postings and Technology Trend Tables